



GLA UNIVERSITY, MATHURA

Department of Computer Engineering Application

SYNOPSIS

on

Airline Reservation System

for

B.Tech CSE 3rd Year

Submitted by

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Abstract

The Airline Reservation System is a web-based application developed to automate the process of airline ticket booking and management. The system enables users to search available flights, check seat availability, book tickets, cancel reservations, and manage bookings online. It eliminates manual errors and reduces the time required for reservation processes.

The project is developed using MERN Stack technologies along with MySQL for relational database management. React.js is used for frontend development, while Node.js and Express.js handle backend logic and RESTful API creation. MySQL is used to maintain structured data such as passenger details, flight schedules, booking records, and transaction information.

The system ensures secure authentication, efficient database management, and real-time booking updates. The expected outcome is to provide a reliable, scalable, and user-friendly airline booking platform.

Keywords

Airline Reservation System, MERN Stack, Express.js, JavaScript, Node.js, Web Application, Online Booking, Database Management, REST API.

1. Introduction

The Airline Reservation System is designed to provide an efficient and automated solution for airline ticket booking. Traditionally, airline bookings were handled manually through agents or offline systems, which often resulted in delays, booking errors, and inefficient record management.

With the advancement of web technologies and database systems, online booking platforms have become essential. This project aims to develop a web-based reservation system that allows users to book flights conveniently from anywhere.

The system integrates frontend and backend technologies using JavaScript-based frameworks and ensures reliable database management using MySQL.

2. Problem Statement

Manual and outdated airline reservation systems suffer from several challenges:

- Delays in ticket booking
- Errors in seat allocation
- Poor data management
- Lack of real-time updates
- Security risks in handling customer data

There is a need for an automated and secure web-based system that provides real-time booking, efficient database storage, and smooth user interaction.

3. Objectives

The primary objectives of this project are:

- To design and develop a web-based Airline Reservation System
- To implement user registration and secure login functionality

- To allow users to search flights and check seat availability
- To enable ticket booking and cancellation
- To manage flight schedules and booking records using MySQL
- To create REST APIs using Node.js and Express.js
- To ensure efficient database management and secure transactions

4. Literature Review

Various airline booking systems have been developed using web technologies and database management systems. Modern reservation platforms use client-server architecture and RESTful APIs to handle booking operations efficiently.

Research shows that combining frontend frameworks like React.js with backend technologies such as Node.js and Express.js improves performance and scalability. Relational databases like MySQL ensure structured data storage and maintain data integrity.

The proposed system adopts these modern technologies to develop a simplified yet functional airline reservation platform suitable for academic implementation.

5. Proposed Methodology

The project will follow a structured development process:

1. Requirement Analysis
2. System Design and Database Schema Design

3. Backend Development using Node.js and Express.js
4. REST API Implementation
5. Frontend Development using React.js and JavaScript
6. Database Integration using MySQL
7. Testing and Debugging
8. Final Deployment

The system will use CRUD operations for managing flights, passengers, and bookings.

6. System Architecture

The Airline Reservation System follows a three-tier architecture:

1. Presentation Layer – React.js User Interface
2. Application Layer – Node.js & Express.js Server
3. Database Layer – MySQL Database

User requests are sent from the frontend to the backend via REST APIs.

The backend processes the request and interacts with the MySQL database to retrieve or store information.

This architecture ensures modularity, scalability, and security.

7. Implementation Plan

The implementation will be divided into phases:

Phase 1: Requirement gathering and system planning

Phase 2: Database design (MySQL schema creation)

Phase 3: Backend API development

Phase 4: Frontend UI development

Phase 5: Integration and Testing

Phase 6: Documentation and Final Deployment

The project will be completed within the academic semester.

8. Expected Results

The expected results of the project include:

- A fully functional web-based airline booking system
- Secure user authentication and data storage
- Real-time seat availability tracking
- Efficient database management using MySQL
- User-friendly interface for customers and administrators

The system will reduce manual errors and improve booking efficiency.

9. Conclusion

The Airline Reservation System provides an automated solution for airline ticket booking using modern web technologies. By integrating MERN Stack components with MySQL, the system ensures efficient data handling, secure transactions, and scalable architecture.

This project enhances practical knowledge of full-stack web development, REST API creation, and database management.

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